Exercise – Understanding sample selection bias using simulation

1. Aim of the exercise

The aim of this exercise is to illustrate how selection bias generates endogeneity through simulation.

2. Theory

Consider the linear regression model

$$y_i = \beta_0 + \beta_1 x_i + u_i,$$

where y_i is the dependent variable, x_i is an explanatory variable, and u_i contains unobserved determinants of y_i.

Assume that

$$\mathbb{E}[u_i \mid x_i] = 0.$$

By the law of iterated expectations, this assumption implies

$$\mathbb{E}[u_i] = 0.$$

It also implies

$$\mathbb{E}[x_i u_i] = 0,$$

so that the regressor x_i is uncorrelated with the disturbance term u_i. Under this condition, the OLS estimator of β_1 is unbiased and consistent.

Suppose, however, that y_i is observed only for a selected subset of the population. To determine which observations are observed, define the latent selection index

$$s_i^* = x_i + u_i.$$

The corresponding selection indicator is

$$s_i = \begin{cases} 1, & \text{if } s_i^* > 0, \\ 0, & \text{otherwise.} \end{cases}$$

An observation is therefore included in the sample only when

$$s_i = 1.$$

The observed sample consists of observations satisfying

$$x_i + u_i > 0.$$

Substituting the definition of s_i^* into the selection rule gives

$$u_i > -x_i.$$

This inequality shows that the observations retained in the sample are selected partly on the basis of the regression disturbance u_i. To estimate β_1 consistently by OLS, the exogeneity condition

$$\mathbb{E}[u_i \mid x_i, s_i = 1] = 0$$

must hold in the observed sample. Does this condition hold? Since only observations satisfying

$$u_i > -x_i$$

are retained, the distribution of u_i in the observed sample is truncated. Consequently,

$$\mathbb{E}[u_i \mid x_i, s_i = 1] = E[u_i \mid u_i > -x_i].$$

Because the truncation point depends on x_i, the conditional expectation depends on x_i.

Therefore,

$$\mathbb{E}[u_i \mid x_i, s_i = 1] \neq 0.$$

It follows that

$$\mathbb{E}[x_i u_i \mid s_i = 1] \neq 0.$$

Thus, although x_i and u_i are uncorrelated in the population, they become correlated in the observed sample.

Consider next the OLS estimator,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})y_i}{\sum_{i=1}^{N}(x_i - \bar{x})^2}.$$

Substituting the population model

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

yields

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{N}(x_i - \bar{x})u_i}{\sum_{i=1}^{N}(x_i - \bar{x})^2}.$$

Taking expectations conditional on the observed sample gives

$$\mathbb{E}[\hat{\beta}_1 \mid s_i = 1] = \beta_1 + \frac{\mathbb{E}[(x_i - \bar{x})u_i \mid s_i = 1]}{\mathbb{E}[(x_i - \bar{x})^2 \mid s_i = 1]}.$$

If the exogeneity condition held, then

$$\mathbb{E}[(x_i - \bar{x})u_i \mid s_i = 1] = 0,$$

and OLS would be unbiased. However, because sample selection induces correlation between x_i and u_i,

$$\mathbb{E}[(x_i - \bar{x})u_i \mid s_i = 1] \neq 0.$$

Therefore,

$$\mathbb{E}[\hat{\beta}_1 \mid s_i = 1] \neq \beta_1.$$

The OLS estimator is biased. Furthermore, because the correlation between x_i and u_i persists as the sample size increases, the OLS estimator is also inconsistent. This phenomenon is known as sample-selection bias.

3. Generate data from a simultaneous equations model

Clear the MATLAB environment by removing all variables from memory, clearing the Command Window, and closing any open figures. This ensures that results from previous analyses do not interfere with the current simulation and helps make the code fully reproducible.

```
%% 3. Initialize MATLAB environment

% Clear workspace, command window, and open figures
clear;
clc;
close all;
```

4. OLS estimation of the first structural equation

Next, we specify the parameters of the data-generating process. We define the sample size and set the true values of the intercept and slope coefficients. These true parameter values will be used to generate the data and serve as benchmarks for evaluating the OLS estimates.

```
%% 4. Set data-generating process parameters

% 4.1. Number of observations
N_obs = 1000;

% 4.2. True regression coefficients
B_0 = 0.2;
B_1 = 0.5;
```

5. Sampling distribution of the OLS estimator

We conduct a Monte Carlo simulation to examine how sample selection affects the OLS estimator. For each of 1,000 replications, we generate a sample from a linear regression model with known parameter values and normally distributed regressors and disturbances.

To introduce sample selection, we construct a latent selection index that depends on both the explanatory variable and the regression error. Observations are retained only when the selection criterion is satisfied. Because selection depends on the error term, the resulting sample is no longer randomly drawn from the population.

For each selected sample, we estimate the regression model using OLS and store the estimated slope coefficient. Repeating this procedure many times produces an empirical sampling distribution for $\hat{\beta}_1$. We can use this distribution to investigate how sample selection affects the bias of the OLS estimator.

```
%% 5. Generate the sampling distribution of the OLS estimator

% 5.1. Number of Monte Carlo simulations
N_sim = 1000;

% 5.2. Preallocate storage for OLS estimates
B_hat_1_sim = NaN(N_sim,1);

% 5.3. Generate samples and estimate OLS repeatedly
for i = 1:N_sim
    % Generate regressor vector
    x_0 = ones(N_obs,1);
```

```
    x_1 = random('Normal',0,1,[N_obs 1]);
    X = [x_0,x_1];
    % Generate error term
    u = random('Normal',0,1,[N_obs 1]);
    % DGP of the population model
    y = B_0 * x_0 + B_1 * x_1 + u;
    % Latent selection index to make x_1 and u correlated
    s_star = x_1 + u; % Or, s_star = x_1 .* u
    % Selection indicator
    s = (s_star > 0);
    % Retain only observed observations
    y_obs = y(s);
    X_obs = X(s,:);
    % OLS estimation
    LSS = lss(y_obs,X_obs);
    % Obtain and store the OLS estimates of beta
    B_hat_1_sim(i,1) = LSS.B_hat(2,1);
end
```

6. Bias of the OLS estimator

We evaluate the bias of the OLS estimator using the empirical sampling distribution obtained from the Monte Carlo simulation. First, we compute the mean of the sampled OLS estimates of β_1. We then calculate the bias as the difference between this average estimate and the true parameter value, β_1. If OLS were unbiased, this difference would be close to zero. However, because sample selection depends on both the explanatory variable and the error term, the selected sample is not randomly drawn from the population. As a result, the OLS estimator is expected to be biased, and the estimated bias provides a numerical measure of the impact of sample selection on the estimation of β_1.

```
%% 6. Bias of the OLS estimator

% 6.1. Compute the mean of the OLS estimates
B_hat_1_mean = mean(B_hat_1_sim);

% 6.2. Compute the bias of the OLS estimator
bias = B_hat_1_mean - B_1;
```

7. Plot the sampling distribution of the OLS estimator

We visualize the empirical sampling distribution of $\hat{\beta}_1$ using a kernel density estimate based on the Monte Carlo replications. The resulting density illustrates how the OLS estimator varies across repeated samples subject to sample selection.

To evaluate the presence of selection bias, we plot both the mean of the sampling distribution and the true parameter value, β_1. If OLS were unbiased, the sampling distribution would be centered around the true parameter value. Any systematic difference between the two indicates bias.

Figure 1 illustrates the effect of sample selection on the OLS estimator. By comparing the

center of the sampling distribution with the true parameter value, we assess both the direction and magnitude of the bias induced by non-random sample selection.

```
%% 7. Plot the sampling distribution of the OLS estimator

% Create a figure
figure('Position',[100 100 1000 1000]);
% Estimate the density
[f,x] = ksdensity(B_hat_1_sim);
% Plot the density
plot(x,f,...
     'Color',[0.000 0.000 0.000], ...
     'DisplayName','Sampling distribution of Beta\_hat\_1')
% Expand the x-axis range
lower = min([x(:);B_1]) - 0.1;
upper = max([x(:);B_1]) + 0.1;
xlim([lower upper])
hold on
% Add the mean of the sampling distribution
line([B_hat_1_mean B_hat_1_mean],ylim,...
     'Color',[1.000 0.000 0.000],...
     'DisplayName','Mean of Beta\_hat\_1')
% Add the true value
line([B_1 B_1],ylim,...
     'Color',[0.000 0.000 1.000],...
     'DisplayName','True value of Beta')
title(['Fig. 1. Sampling distribution of the OLS estimator ' ...
    'under sample selection'])
legend('show')
xlabel('Beta\_hat\_1')
ylabel('Density')
hold off
```

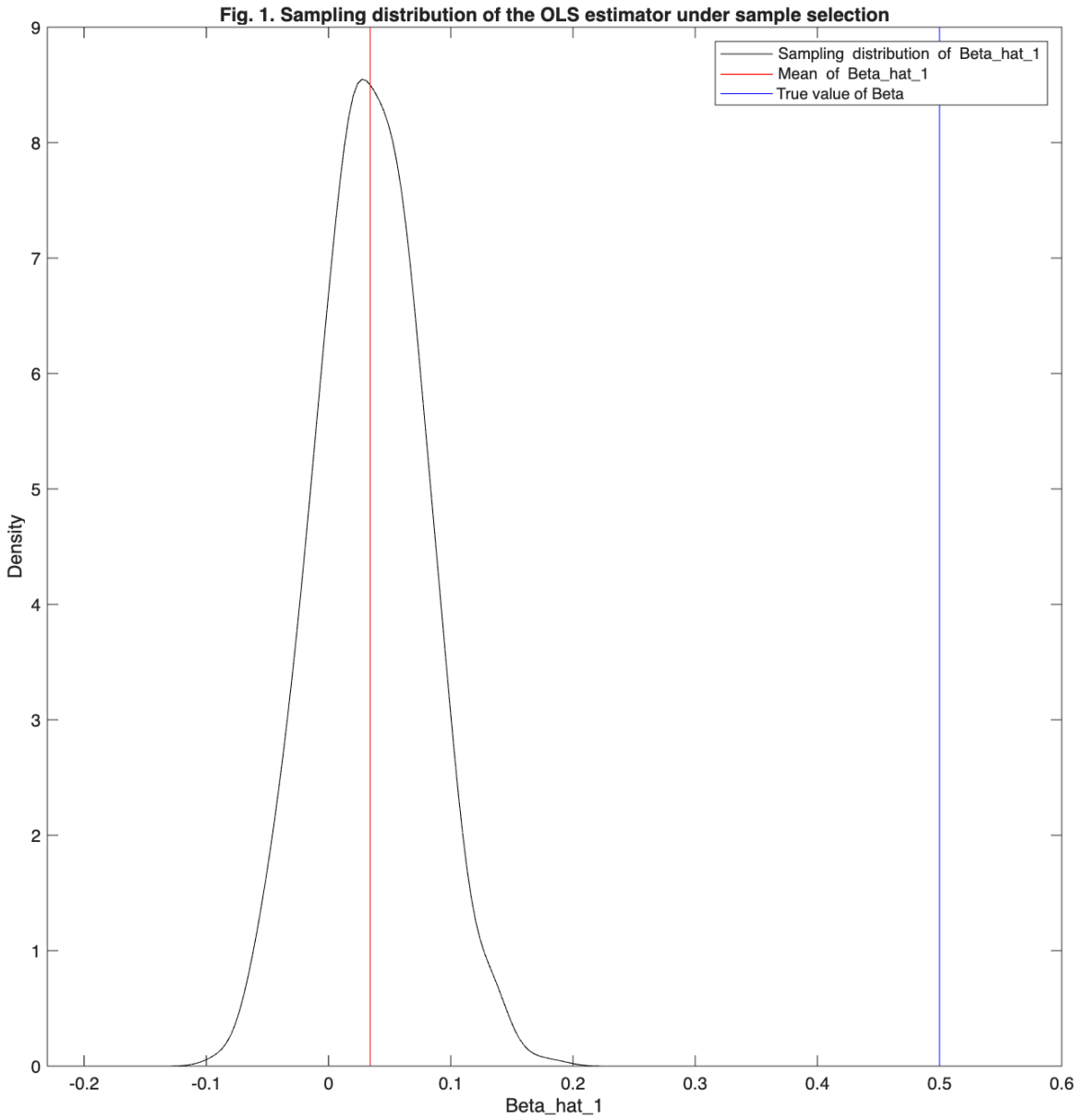


Figure 1: Sampling distribution of the OLS estimator under sample selection.

9 . Final notes

This file is prepared by Tunga Kantarcı. This file and the accompanying MATLAB files are available on GitHub and can be accessed via this link.